AIM

To study and implement visualization techniques for high-dimensional data using Parallel Coordinates Chart, Deviation Chart, and Andrews Curves.

OBJECTIVES

- Understand the challenges of visualizing data with many features (dimensions).
- Implement and interpret Parallel Coordinates Chart for multi-dimensional data.
- Implement and interpret Deviation Chart to analyze data spread from a reference.
- Implement and interpret Andrews Curves for pattern discovery in high-dimensional datasets.
- Compare the insights offered by each visualization technique.

THEORY

High-dimensional data refers to datasets that contain a large number of features or variables (dimensions). When the number of dimensions exceeds three, direct visual representation becomes impossible. Specialized visualization techniques are used to project, summarize, or map multi-dimensional data into a 2D or 3D space for human interpretation.

1. Parallel Coordinates Chart

A Parallel Coordinates Chart (also called a Parallel Coordinates Plot or PCP) is a technique for visualizing multivariate numerical data. Instead of using orthogonal axes, it arranges all axes in parallel (vertically), equally spaced horizontally. Each data record (observation) is represented as a polyline that crosses each axis at the value corresponding to that variable.

Key Characteristics:

- Each vertical axis represents one variable/feature of the dataset.
- Each data point is a connected line crossing all axes.
- Clusters appear as bundles of parallel or crossing lines.
- Correlations between adjacent axes are visible as crossing patterns (negative) or parallel patterns (positive).

Advantages:

- Handles datasets with many dimensions without loss of information.
- Useful for detecting clusters, outliers, and variable correlations.
- Supports interactive filtering by selecting ranges on individual axes.

Limitations:

- Can become cluttered and hard to read with many data points.
- Correlations are only clearly visible between adjacent axes.
- Ordering of axes significantly affects interpretation.

Python Implementation (using pandas/matplotlib):

```
from pandas.plotting import parallel_coordinates import matplotlib.pyplot as plt
import pandas as pd  df = pd.read_csv('dataset.csv') plt.figure(figsize=(10, 6))
parallel_coordinates(df, 'class_column', colormap='Set1') plt.title('Parallel
Coordinates Chart') plt.xlabel('Features') plt.ylabel('Values')
plt.legend(loc='upper right') plt.tight_layout() plt.show()
```

2. Deviation Chart

A Deviation Chart visualizes how individual data values deviate from a reference point — typically the mean, median, or a benchmark value. It highlights how much each observation (or category) is above or below the central value, making it easy to identify outliers and asymmetry in data distributions.

Key Characteristics:

- Deviation is computed as: Deviation = Observed Value − Reference Value (e.g., mean).
- Positive deviations extend in one direction; negative deviations extend in the opposite direction.
- Commonly displayed as diverging bar charts or lollipop charts.

Use Cases:

- Comparing feature values relative to the dataset mean (z-score visualization).
- Identifying features that are anomalously high or low for a given observation.
- Useful in fraud detection and anomaly analysis case studies.

Python Implementation:

```
import matplotlib.pyplot as plt import pandas as pd import numpy as np  df =
pd.read_csv('dataset.csv') features = df.select_dtypes(include=np.number).columns
means = df[features].mean() row = df[features].iloc[0]  # Select a sample row
deviations = row - means  colors = ['steelblue' if d >= 0 else 'tomato' for d in
deviations] plt.figure(figsize=(10, 6)) plt.barh(features, deviations,
color=colors) plt.axvline(0, color='black', linewidth=0.8) plt.title('Deviation
Chart (from Mean)') plt.xlabel('Deviation') plt.tight_layout() plt.show()
```

3. Andrews Curves

Andrews Curves (proposed by D.F. Andrews, 1972) represent each multivariate data point as a Fourier series (a smooth curve) over the interval [−π, π]. Each dimension of the data contributes to the amplitude of the corresponding harmonic in the Fourier function. Data points that are similar in their original feature space will produce similar curves, while dissimilar points produce different curves.

Mathematical Basis:

$$f(t) = x_1/\sqrt{2} + x_2 \sin(t) + x_3 \cos(t) + x_4 \sin(2t) + x_5 \cos(2t) + \ldots$$

Where x_1, x_2, x_3, … are the feature values of a data point and t ranges over [−π, π].

Key Characteristics:

- Each data point becomes a continuous function (curve) over the t-axis.
- Similar data points produce similar-looking curves that cluster together visually.
- Preserves distances: if two points are close in feature space, their Andrews curves will be close.
- The first feature (x_1) has the strongest influence on the shape of the curve.

Advantages:

- Effective for discovering clusters and groupings in high-dimensional data.
- Smooth curves are visually easier to distinguish than parallel line plots.
- Statistically grounded representation (Fourier series).

Limitations:

- The importance of features changes with their ordering in the dataset.
- Difficult to interpret individual feature contributions from the plot alone.

Python Implementation:

```
from pandas.plotting import andrews_curves import matplotlib.pyplot as plt import
pandas as pd  df = pd.read_csv('dataset.csv') plt.figure(figsize=(10, 6))
andrews_curves(df, 'class_column', colormap='tab10') plt.title('Andrews Curves')
```

```
plt.xlabel('t') plt.ylabel('f(t)') plt.legend(loc='upper right') plt.tight_layout()
plt.show()
```

COMPARISON OF TECHNIQUES

Criteria	Parallel Coordinates	Deviation Chart	Andrews Curves
Primary Use	Multi-variable comparison across records	Identifying deviation from reference/mean	Cluster and pattern discovery
Visual Form	Polylines across parallel axes	Diverging bar/lollipop chart	Smooth Fourier-based curves
Handles Many Dims?	Yes	Yes	Yes
Cluster Detection	Moderate	Low	High
Outlier Detection	High	High	Moderate
Correlation Visible?	Yes (adjacent axes)	No	Indirectly
Python Library	pandas.plotting	matplotlib	pandas.plotting

DATASET USED

For this assignment, the Iris dataset is used as a sample to demonstrate all three visualization techniques. The Iris dataset contains 150 observations across 4 features (sepal length, sepal width, petal length, petal width) and 3 class labels (Setosa, Versicolor, Virginica).

- Source: UCI Machine Learning Repository / sklearn.datasets
- Number of Records: 150
- Number of Features: 4 numeric features + 1 class label
- Classes: Iris-setosa, Iris-versicolor, Iris-virginica

OBSERVATIONS

- Parallel Coordinates revealed clear separation between the Iris-setosa class and the other two classes across all four features.
- Deviation Chart for individual observations highlighted which features deviated significantly from the dataset mean, useful in anomaly detection scenarios.
- Andrews Curves showed tight clusters for Iris-setosa and overlapping curves for Versicolor and Virginica, confirming their similarity in feature space.
- All three techniques successfully visualized the 4-dimensional dataset without losing class structure information.

CONCLUSION

This assignment demonstrated three important high-dimensional data visualization techniques. Parallel Coordinates Charts are ideal for comparing multiple variables simultaneously across all records. Deviation Charts are most effective for spotting outliers and understanding how individual data points compare to the overall distribution. Andrews Curves provide a mathematically grounded method for detecting clusters and patterns in high-dimensional feature spaces. Each technique has its own strengths, and the choice depends on the specific analysis goal and dataset characteristics.